

Upward and Downward Causation from a Relational-Horizontal Ontological Perspective

Gil C. Santos

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Abstract Downward causation (DC) exercised by emergent properties of wholes upon their lower-level constituents' properties has been accused of conceptual and metaphysical incoherence. Only upward causation is usually peacefully accepted. The aim of this paper is to criticize and refuse (1) the traditional *hierarchical-vertical* way of conceiving both types of causation, although preserving their deepest ontological significance, as well as (2) the widespread acceptance of the traditional *atomistic-combinatorial* view of the entities and the relations that constitute the so-called 'emergence base'. Assuming those two perspectives with no reserves, we are condemned to confine our debate to the question of whether reified *wholes* can have the power to downwardly change or influence their lower-level parts, a question which seems profoundly misleading to me. I therefore propose an alternative relational ontological view, assuming a straightforward *horizontal* and *intra-level* way of representing those putative cases of cross-level causation. I finally confront two recent replies to Kim's well-known objections to DC—Craver and Bechtel (Biol Philos 22:547–563, 2007) and Kistler (Philos Psychol 22(5):595–609, 2009)—, emphasizing their global positive approaches, as well as the reasons why their accounts still seem insufficient to me. I conclude arguing that both Kim's principle of the causal closure of the physical domain and its allegation of an overdetermination in cases of DC can be surpassed by the new relational ontological perspective presented here.

Keywords Downward causation · Upward causation · Atomism · Qualitative change · Relational ontology

G. C. Santos (✉)

Centro de Filosofia das Ciências da Universidade de Lisboa, Universidade de Lisboa, Lisbon,
Portugal
e-mail: gilcosan@gmail.com

1 Kim's Objections to Downward Causation

The problem of downward causation (DC) is highly relevant to Philosophy of Science because the existence of ontological emergent phenomena occurring at new qualitative levels of organization of reality, and possessing causal powers that can affect behaviors at lower levels, renders the widespread micro-reductionist or microphysicalist ontological and epistemological claims unjustifiable; such as the principle of the causal closure of the physical domain and the complete explanatory reducibility (at least, in principle) of all levels of organization in terms of the laws of physics.

Two basic features define putative examples of ontological emergent phenomena: they are qualitative novelties (different from the properties from which they emerge) and they manifest some degree of causal autonomy over and above the set of their base elements. Thus, ontological emergent phenomena must bring distinctive causal powers with them, which contribute to the evolving causal structure of the world and that causally influence processes at lower levels (Kim 2006: 557).

As it is widely known, Jaegwon Kim has accused the notion of DC of metaphysical and conceptual incoherence, which implies a potential circularity. This critique was elaborated based on the assumption of two principles:

1. The principle of the 'causal closure of the physical domain's states that, for a given physical event e that takes place at time t , for each time t^* preceding t , there is a complete physical cause c (at t^*) of e (Kim 1998: 37–38).
2. The principle of 'causal/explanatory exclusion' (non-overdetermination) states that if an event e at t_2 has a complete physical cause c at time t_1 , then, in addition, it does not have another complete cause d at the same time t_1 , which is independent of c . As Kim says, "there can be no more than a single complete and independent explanation of any one event" (Kim 1988: 233).

From the intersection of these two principles Kim infers some of its major objections to DC:

1. "how [can] properties or phenomena have causal effects on the very processes that make their emergence possible and without which they could not even exist?" (Kim 2000: 305);
2. "If an emergent, M , emerges from basal condition P , why can't P displace M as a cause of any putative effect of M ? Why can't P do all the work in explaining why any alleged effect of M occurred?" (Kim 1999: 32).

Several replies to Kim's approach have been developed, but, to the best of my knowledge, the cornerstone and most problematic assumptions of the underlying metaphysics of Kim's arguments are yet to be revealed. This is what I intend to do now by addressing Kim's notion of the 'emergence base'.

1.1 The Emergence Base and the Atomistic Metaphysics

According to Kim, bottom-up determination means that the micro-properties and relations of parts determine the macro-properties and behaviors of wholes. This is the ‘emergence base’: a set of parts with their own properties and relations. But how should we conceive this emergence base? We can start to address this question by identifying and evaluating two different ways of conceiving the nature of the individuals that constitute the emergence base, and the nature of their relations:

1. Either we conceive the individuals as endowed with a set of absolutely intrinsic (monadic) and qualitatively immutable defining properties, their relations being therefore considered to be mere arrangements or combinations, only affecting quantitatively those intrinsic properties (in such a case the relations only tell us how the elements are combined and how the set or the whole is configured);
2. Or we admit that entities are not qualitatively immutable, being able to change their properties (loosing and acquiring new qualities), and, in this sense, their relations, which apart from the power to generate the global configuration of sets of elements, also play a direct causal role in the way component parts behave and in the properties they will manifest (in such case, relations have the power to construct and transform the qualitative identities of their relata).

For the sake of precision, I wish to make clear that by intrinsic properties I mean the properties that each entity has of itself, despite its extrinsic relations with other entities, and relational properties are properties that each entity has and acquires due to its extrinsic relations with other entities in its environment. Obviously, not all relations reach the point of producing a *qualitative* kind of change in their relata, and, indeed, in some cases we can explain a given phenomenon in terms of a certain set of compositional or additive relations among its different components or causes, through which the relata maintain their qualitative identities intact, being affected only in a quantitative way, that is, merely varying the magnitudes and proportions of their intrinsic properties. Nevertheless, there are also cases where these relations qualitatively change the identities and behaviors of the relata.

Following the first interpretation, we adopt an Atomistic metaphysical view. According to this concept, all entities, beginning by the most elementary building blocks of the world (whatever they are: particles or waves, tropes, substances, states of affairs, etc.) are taken to be a self-sufficient kind of entity endowed with a set of absolutely intrinsic defining properties that are qualitatively immutable through their extrinsic relations. Each entity is thus taken as a preexisting unity of wholes and relational structures. Therefore, all relations are seen as mere forms of combinations, arrangements, or spatial configurations that keep the qualitative identities of their relata intact, being only able to change quantitatively the degree, magnitude or proportions of the relata’s pre-existing properties. This is indeed the main ontological tenet of Atomism: no elementary entity or compound system can

qualitatively change its identity, that is, acquire and lose properties, through their extrinsic relations (Guthrie 1965; Curd 2002).¹

Kim adopted this atomistic way of thinking, since its arguments are framed within the supervenience theory, which is itself a derivative product of an atomistic view of the ontological nature of particulars and relations. This is the reason why his supervenience theory was qualified as mereological:

[S]upervenience theses, when applied to the layered model, turn into claims of mereological supervenience, the doctrine that properties of wholes are fixed by the properties and relations that characterize their parts. A general claim of macro–micro supervenience then becomes the *Democritean atomistic doctrine* that the world is the way it is because the microworld is the way it is. (Kim 1998: 18).

From this follows Kim’s conceptual scheme of analysis of wholes:

any physical system can be exhaustively described in terms of (i) the basic particles that constitute it (...); (ii) all the *intrinsic* properties of these particles; and (iii) the relations that configure these particles into a structure (...). Such a description will give us the total “relatedness” of basal constituents. (Kim 1999: 6–7).

2 A Relational Ontology: An alternative View

The first possible interpretation, as it was shown, is based on an Atomistic Ontology. Let us now consider the second possible interpretation—a Relational Ontology—according to which relations are not mere configurations or combinatorial

¹ Greek atomists, as Empedocles and Anaxagoras, tried to reconcile the new Eleatic theory of Being with the need to rehabilitate the study of the pluralistic ontology of physical reality. Indeed, while for all the Ionian physicists, ‘being’ designates the plurality of singular things (*ta onta*) that the human experience captures, in Parmenides the notion of Being will be expressed as singular (*to eon*), referring to Being in general, as an absolute unity, homogeneous, indivisible, timeless, motionless and unchanging. As it is well known, Atomism advocates the impossibility of both qualitative and substantial kinds of change of the most elementary entities of physical reality (the atoms conceived into the light of Parmenides’ theory of Being), sustaining that all non-fundamental entities and properties arise in virtue of (supervening upon) mere additive, aggregative or combinatorial relations among the atoms that leave intact their own qualitative identities (see Wardy 1988; Graham 1999). If one wants to look for an opposite perspective in the Greek philosophy itself is enough to recall the Aristotelian ontology, according to which the most basic entities of physical reality (the four elements) undergo not only quantitative, but also qualitative and substantial kind of transformations through their own interactions. Lastly, we may observe that the mechanistic paradigm of modern physics is the direct expression of this metaphysical atomism. As David Bohm once wrote, “the most essential and characteristic feature of mechanism (...) [is] to reduce everything in the whole universe completely and perfectly to purely quantitative changes in a few basic kinds of entities (in this case, bodies and fields, or fields alone, as in the point of view of Einstein), which themselves never change qualitatively” (Bohm 1984: 47). Indeed, “[a]t bottom, the only changes that are regarded as possible within this scheme are quantitative changes in the parameters or functions (...), while fundamental qualitative changes in the modes of being of the basic entities and in the forms in which the basic laws are to be expressed are not regarded as possible. Thus, the essence of the mechanistic position lies in its assumption of fixed basic qualities, which means that the laws themselves will finally reduce to purely quantitative relationships.” (*idem*: 131).

arrangements between qualitatively unchanging entities, but are instead capable of constructing and transforming the qualitative identities, behaviors and causal powers of their relata. According to this alternative view, the ‘emergence base’ is viewed as a set of entities able to undergo qualitative change through their own intra-level relations. In this sense, relations play a direct causal role in the way entities behave, influencing and qualitatively changing them.

Now, if we assume the possibility that entities can change their properties through their relations, and thus acquire new properties and causal powers within some specific relational contexts (Santos 2014), then, the emergent properties that arise at the level of the systems taken as wholes will be the direct ‘expression’ of those very sets of transformative interactions (Schröder 1998: 447), being directly explainable in their own terms.

In such a case, an emergent systemic property cannot be micro-explainable in terms of the standard atomistic micro-reductionist model, that is, it cannot be derived:

1. “from an exhaustive knowledge of the non-relational properties of the parts”, in addition to
2. “the lower-level laws that obtain over these properties”;
3. “to laws of composition that obtain among lower-level entities (e.g., additivity, fundamental forces)”;
- and
4. to “statements of definition” (Garson 2006: 230; Hüttemann 2004: 35).²

In this sense, an emergent macro-systemic property can only be directly explained in terms of the set of the qualitatively transformative interactions between the lower-level constituents, and in terms of the new law-like generalizations that describe the correlations of the new component’s relational properties, accounting for the new patterns of causal behaviors that they exhibit. A new level of causation and of explanation would thus arise above the set of the intrinsic (monadic) properties of the system’s constituents and their specific low-level laws and theories. As I argued before (Santos 2014), the qualitative change of the micro-components within a given compound system through their own intra-level relations is the only way to justify the inadequacy of micro-reductionism, and correlatively, to demonstrate the ontologically emergent nature of a system’s macro-properties.

If we admit the ontological possibility that entities, as relata of relational contexts or as component parts of wholes, can influence and change each other through their own intra-level relations, we no longer need to reify wholes in order to attribute them the power of interacting downwardly with their parts.

² According to the most refined version of this micro-explanatory model of reduction presented by Hüttemann (2004: 35), such a reduction needs three types of ingredients: (1) general laws concerning the behavior the component parts considered by themselves as units of organization (thus, considered in a ideal isolation); and (2) general laws of composition (for vectorial, scalar or tensorial magnitudes); and (3) general laws that account for the mechanist interactions among the lower-level parts that merely account for the quantitative co-variations of their intrinsic properties (such as ‘mass’, ‘position’, ‘motion’, ‘velocity’, ‘acceleration’, etc.).

My thesis is that philosophers have been driven to debate whether *wholes* as reified entities can have the power to downwardly cause or constrain their parts' behaviors, because they have been tacitly assuming, right from the start, that parts could not perform this task through their own interactions. That is, philosophers have been accepting (more, or less, unconsciously) an atomistic view of entities' natures.

On the contrary, in a relational ontology, it makes no sense to assume that one special kind of entity called whole is causally interacting with another kind of entity called part, sending *downwardly* and *upwardly* causal messages to each other.

From a relational ontological viewpoint, 'whole' is just a *word* that names the relational totality of the individual *relata* and their relations, the only real causal agents being the *relata* and their intra-level relations. Therefore, if we are willing to defend ontological emergence and some version of DC we have to be ready to reconsider and re-conceive:

1. the nature of the individuals: are they like metaphysical atoms, endowed with a set of absolutely intrinsic (monadic) defining properties and unable to change their quality through their extrinsic relations?; and
2. the nature and causal power of their relations: are relations mere configurative and combinatorial kinds of arrangements, only able to quantitatively change the magnitude and proportion of the pre-given intrinsic properties of their *relata*?

From the point of view of a relational ontology, entities undergo qualitative changes and indeed construct their identities through their intrinsic and extrinsic relations. In this sense, we abandon the hierarchical and vertical layered model of reality where wholes (as relational systems) exist above the level of its parts (as *relata*) and their relations. This hierarchical view is a natural way of representing the different levels of composition and organization, but it is an inadequate and in fact misleading model of representation when we deal with *causality*.³

Therefore, we can preserve the meaning of both upward (UC) and DC without assuming the existence of causes that “go up” and “go down” between parts and wholes. UC and DC must be conceived as two different aspects of the *same ongoing intra-level process of systemic relational causation* between different *relata* that belong to the same relational domain. UC and DC name two different types of changes produced by the same set of causal relations whose effects are distributed and propagated across an entire relational system, through the intra-level relational processes between the *relata*:

³ It is worth to observe that the hierarchical-vertical representation in terms of substantive parts and wholes is far from being the only one way of representing the natural world. The developmental ecologist Stanley N. Salthe, for example, has distinguished two different types of hierarchy, which have been used in models of natural phenomena, including the biological, namely the scalar or composition hierarchy, and the specification or subsumption hierarchy (see Salthe 2009, 2012). In the present case, I want to emphasize the difference between representing relations of composition and organization, on one hand, and causal relations, on the other hand. For an analysis of the theory of levels in terms of its ontological significance and major philosophical implications, see Poli (2001).

1. UC names the global effect of the relata's set of *individual* behaviors that jointly cause (construct) the identity and the behavior of their relational system taken as a whole. Therefore, UC refers to the set of causal contributions of each relatum to the overall behavior of its system.
2. DC names the global effect that the set of qualitatively transformative and coercive systemic interactions exercises upon the individual behaviors of each relatum—in short, DC refers to the way that each relatum is *qualitatively affected* by all other relata's actions and interactions.

This latter kind of causal change can only metaphorically be called higher-level because it somehow 'transcends' the individual behaviors of each relatum taken by itself. Still, both UC and DC are intra-level, since relational causation is just the joint effect of a set of transformative, coercive and co-dependent interactions that individuals, as relata of a common relational system, have with each other. If different agents start to interact coercively, influencing and changing each other's behavior—thus acquiring new properties, behaviors or causal powers—new relational patterns of co-dependent causal behaviors (able to be described by new nomological generalizations) will be diffused, disseminated and distributed across the entire relational system, affecting the behaviors of each individual relatum.

Suppose that some entities ($a, b, c \dots$) are involved in some system of relations (R_i). The idea is that those entities would not behave the way they do if they were not embedded, as interactants, in that relational system (R_i). This is what the notion of DC always intended to capture, but now in a *relational-horizontal* way of representing systemic causation. UC and DC are thus to be conceived as just two different aspects (conceptually distinguished by a process of abstraction) of the same ongoing intra-level process of relational causation among a set of relata that co-construct each other's identities and behaviors within a common relational system.

2.1 A Graph Representation

A graph representation can give us a nice relational-horizontal way of representing systemic causation. Instead of the standard vertical representations of UC and DC with arrows pointing to a top level and to a bottom level, we propose a *horizontal* model of representation (Fig. 1). Consider the following graph representation of the relational system S with six subsystems as relata (A, B, C, D, E, F):

A whole is represented as a graph conceived as a pair $S = (V, E)$, where the elements of V are the *relata* represented as vertices or nodes, and the elements of E are their relations represented as edges or lines. Now, imagine that this system S is a living cell and its relata (A, B, C, D, E, F) are the cell's components: DNA molecules, RNA molecules, proteins, ribosomes, etc. The whole called 'cell' names (is nothing but) the relational system of those relata and their relations, and although the relata are represented here as nodes, they must of course be seen, in themselves, as subgraphs, that is, as having their own internal relational structures.

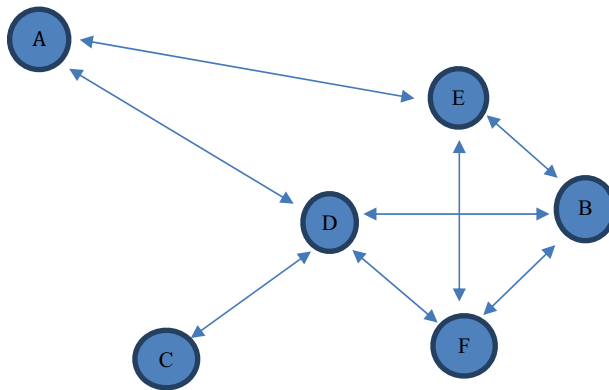


Fig. 1 Graph representation of a whole represented as a relational system

In light of such representation, UC is re-conceived as the set of the individual causal contributions exercised by each interactant upon all other relata and their interactions, and DC is re-conceived as the set of the transformative and coercive relations felt by each interactant in virtue of being directly and indirectly affected by all the relations and new causal patterns of behaviors distributed across the entire relational system. For example, the interactant *E* is causally contributing to the construction of the overall identity and behavior of the relational system through its own actions and interactions and, at the same time, is *directly* influenced by its relations to A, B, and F, and is *indirectly* influenced by the relations that A, B, and F have with their own direct relata D and B—more precisely, it is indirectly affected by the continuous outcomes of the relations A/D; F/D; F/B. This process of causal interconnectedness can be further analyzed by identifying the causal effects directly and indirectly felt by D until we cover the whole system's relata and relations, and in this way capturing the entire causal network *S*.

In this relational-horizontal way of representing 'wholes' and 'parts' we can capture everything that is meant by DC and UC without the need of assuming the existence of causes "going up" and "going down", with wholes somehow living on the top floors and parts lying on the floors below. As it was said, the only causal agents are the relata and their relations, the whole being merely the overall or net effect of those intra-level activities.

According to the graph theory, our cell (the system *S*) is represented as a directed graph, the interactants (A, B, C, D, E, F) being represented as vertices (or nodes), and their mutual interactions being represented by arcs, as biconnected and oriented edges, since they represent relations exercised in both directions. UC and DC are thus represented as the different and complementary horizontal directions in the graph's paths (Rosenstiehl 1979; Diestel 2000).

Now, let us suppose that *S* represents a living cell, the nodes representing its constituent parts. As the Harvard geneticist Richard Lewontin has strongly emphasized, a DNA molecule is among the most nonreactive, chemically inert molecules in the living world. Therefore, DNA has no power to reproduce itself and

by itself it does not make proteins or anything else relevant. A segment of DNA isolated from the cytoplasmic machinery of ribosomes and proteins has no magic power of self-reproduction (Gray 2001: 193). But when DNA molecules interact with other kinds of relata, such proteins, and RNA molecules, they get to be copied due to their transformative interactions with the cytoplasmic machinery of ribosomes and the proteins:

How is that a mere molecule can have both the power of self-reproduction and self-action, being the cause of itself and the cause of all other things? (...) First, DNA is not self-replicating, second, it makes nothing, and third, organisms are not determined by it. DNA molecule is a dead molecule, among the most nonreactive, chemically inert molecules in the living world. (...) DNA has no power to reproduce itself. Rather, it is produced of elementary materials by a complex cellular machinery of proteins (...) No living molecule is self-reproducing. Only whole cells may contain all the necessary machinery for 'self'-reproduction and even they, in the process of development, lose that capacity (Lewontin 2000: 141–142).

Now, are we willing to say that DNA molecules are downwardly affected by the whole called cell? Well, only for the sake of brevity. We may want to save time avoiding the need to specify all the details associated with the set of transformative interactions among the intra-level relata A, B, C, D, E, F... Still, the whole called 'cell' is nothing but this organized network of causal transformations among those kinds of relata.

To say that the causal power once attributed to the wholes (e.g. the cells) can now be attributed to the intra-level causal relations among the intracellular agents does not mean that the so-called high-level theory of the cells can be reducible to the low-level theory of the cell's components (e.g. the molecules). In fact, it is precisely the opposite: contemporary biologists that criticize the central dogma of molecular biology (the genetic dogma) emphasize that we should not stop at the level of the Molecular Genetics and Molecular Biology, focusing on the mere *intrinsic* properties of the micro-constituent parts, but rather focus our research on Cell Biology—that is, we should study DNA molecules and genes in their interactions with other agents or relata such as RNA molecules, cytoplasm, proteins, and ribosomes.

DNA molecules are not self-sufficient entities endowed with all their causal powers *ab initio*. Thus, no theory of their intrinsic (monadic) defining properties in addition to some laws of mere composition (the project of micro-explanatory reductionism) can account for the (new) types of behaviors and causal powers that DNA molecules manifest *when* they interact in the right way with the right kinds of relata. The known causal powers of DNA molecules only exist in virtue of their transformative interactions with RNA molecules, proteins and ribosomes within the specific relational domain that we call a cell (see Atlan 1999; Atlan and Koppel 1999; Oyama et al. 2001).⁴

⁴ Griffiths and Gray (1994: 299): "how exactly is it that has the power to replicate itself? A segment of DNA isolated from the cytoplasmic machinery of ribosomes and proteins has no such power".

In fact, according to the relational ontological perspective that we defend here, the existence-conditions, the identity and the causal behavior of any entity must always be conceived and explained as constructed by the interplay of its intrinsic and extrinsic relational processes. I thus advocate a relational ontology in analogous terms of the ‘constructivist interactionism’ proposed by Susan Oyama (2000) in the study of biological and cognitive development and evolution, according to the new developmental systems approach. From this perspective, known as the Development Systems Theory, it is no surprise that the causal privilege that is commonly given to genes is criticized, since in the production of an organism should not regard “the internal and the external, genes and environment, as alternative causes”, denying “the claim of the ontologically independent status of the causes as causes, aside from their interaction in the effects produced”.⁵

In sum, from a relational ontological approach, entities’ causal powers, capacities or properties can be acquired and lost, that is, entities can undergo processes of qualitative changes through their own intra-level relations, and this is everything that is needed in order to re-conceive DC in terms of a straightforward horizontal perspective.

3 Craver and Bechtel’s View and Kistler’s Counterproposal

Craver and Bechtel 2007’s essay, as well Kistler 2009’s reply to it, represent a first step towards the relational perspective that I have been defending here. Still there are as well some fundamental problems in their accounts that I would like to point out in order to distinguish and clarify my own view.

These authors want to defend DC and, at the same time, argue that causal relations are exclusively intra-level. I fully agree with them on that, as it is clear from my previous analysis.

Craver and Bechtel (2007: 551–555) defend that any *inter*-level causal claim must be analyzed in two components or dimensions:

1. a *causal* dimension consisting in the cause-effects intra-level relationships of all component parts of a mechanism; and
2. a *constitutive* (non-causal) dimension consisting in the fact that since each mechanism is made up of its parts, the activities of each part do not need to be

⁵ As Lewontin says in the ‘Foreword’ to the second edition of Oyama’s book, ‘constructivist interactionism’ is the view that “in the production of an organism one cannot regard the internal and the external, genes and environment, as alternative causes”, denying “the claim of the ontologically independent status of the causes as causes, aside from their interaction in the effects produced”; indeed, “they become causes only at their nexus, and they cannot exist as causes except in their simultaneous action” (Oyama 2000: 14–15). According to the constructionist, developmental systems perspective, traits are not transmitted to offspring, but are instead made or reconstructed in development. Likewise, evolution is not a matter of organisms or populations being molded by their environments, but of organism-environment systems reciprocally changing over time. Therefore, “the life cycle of an organism is developmentally constructed, not programmed or preformed. It comes into being through interactions between the organism and its surroundings as well as interactions within the organism.” (Oyama et al. 2001: 4). See also Gray (1992).

transmitted upward, nor the activities of the mechanism as a whole need to be transmitted downward, to have their ‘effects’; that is, the effects are inherited constitutively, not causally.

In short, inter-level relationships are not causal relations, but *constitutive* kind of relations. Inter-level relationships are “mechanistically mediated effects”, that is, “hybrids of causal and constitutive relations” (Craver and Bechtel 2007: 562). Therefore, in putative cases of DC what we have is a constitutive relationship. Either the component parts contained in a higher-level system are simply “carried along for the ride”, or “the lower-level constituents are enlisted in the activities of the higher-level mechanism” (Craver and Bechtel 2007: 558):

As the mechanism as a whole is put into new conditions, it is organized such that its components change with those conditions. Still the same strategy works: In each case, the putative inter-level claim is analyzed into a causal claim coupled with one or more constituency claims (Craver and Bechtel 2007: 561; see also Bechtel 2008: 153–155).

Now, Kistler (2009: 596) agrees with Craver and Bechtel that “inter-level causal claims can be analyzed in a causal and a non-causal part”, but disagrees “about their interpretation of the non-causal part” (Kistler 2009: 596). According to Kistler (2009: 601), constitution cannot be the relation that bridges between levels because ‘constitution’ is *not* identity, being therefore an *asymmetric* relationship that cannot account for DC: “top-down determination cannot be analyzed in terms of constitution” (Kistler 2009: 600).

Craver and Bechtel indeed assume that constitution is a symmetrical relationship, because “a change in the parts is manifest as a change in the mechanism as a whole, and a change in the mechanism is also a change in at least some of its component parts” (Craver and Bechtel 2007: 554).

Nevertheless, Kistler denies this. First, if A constitutes B, it is impossible that B constitutes A. Second, some systems or mechanisms allow for “multiple constitution” (Kistler gives the example of a cloud and its collection of drops). Therefore, the persistence and evolution of a system cannot be accounted by the set of its constituents since they can change through time. In short, “the collection of drops constitutes the cloud but the cloud doesn’t constitute the collection of drops. Second, a given object can be, successively or alternatively, constituted by more than one collection of parts. One might express this by saying that some objects allow for ‘multiple constitution’.” (Kistler 2009: 601).

Driven by this line of argumentation, Kistler proposes to substitute the non-causal inter-level relationship of constitution, by the non-causal relationship of *constraint*. Accordingly, DC is the *partial* coercive power exercised by the whole on its component parts, reducing the *degrees of freedom* of their own intrinsic properties. In Kistler’s own words,

Each constraint on a macroscopic system diminishes the number of possible states of its constituents. However, as long as there are fewer constraints than degrees of freedom, the constraints on a system determine its state, and the

state of its parts, only partially. Contrary to constitution, constraint is not an asymmetric relation. One can say that the state of the parts of a system constrains the state of the whole; but it can also be correct to say that the state of the whole constrains the states of the parts, as when the temperature or the volume of the container of a gas limits the degrees of freedom of the molecules constituting it (Kistler 2009: 602–603).

Constraints are a form of non-causal determination, according to Kistler, because “if the macroscopic state of a system exercises a constraint on the state of one of its parts, the origin of the constraint (the state of the system) and what it constrains (the state of the part) occupy the same place at the same time” (Kistler 2009: 604). “But—adds Kistler—“cause and effect must not overlap in space, and second, causes must precede their effects. Therefore, constraints do not cause what they constrain” (*idem*). However, these are not sufficient reasons to be skeptical about inter-level determination:

Top-down causation is conceivable if it is understood in terms of the influence of a higher level property C of a whole mechanism at time t_1 on a property e of a lower-level part at some later time t_2 . If a higher-level law imposes a constraint on the state of system S, a higher-level property of S at time t_1 can be (partly) causally responsible for a lower-level property that one of the components of S has at some later time t_2 . (Kistler 2009: 604).

As I said, I fully agree with these three authors, both on their defense of the ontological significance of what the notion of DC intends to express and on their arguments in favor of an exclusive intra-level reading of the notion of causation.

Nevertheless, I have also some disagreements. First, none of the two interpretations acknowledge the fact that relations *qualitatively* change the individual behaviors of the relata. Relations do not merely combine, organize, or co-ordinate the activities or operations of their relata. Second, ignoring the quality transformative power of causal relations within emergent relational systems, they also fail to *distinguish* between the specific kind of systemic causation that takes place in emergent systems from the systemic causation occurring in non-emergent systems but where relations and interactions also take place. Nevertheless, without such a distinction DC would be a trivial phenomenon.

Now, as we already noted, according to Kistler, Craver and Bechtel’s notion of ‘constitutive’ relations only accounts for the inter-level UC, not to DC, since constitution—in contrast with identity—is an asymmetric notion. Constraint—argues Kistler—is a symmetric kind of relation working for both inter-level UC and DC. Still, although ‘constraint’ is conceivable as a non-traditional type of mechanist causation, it seems to me too weak in what concerns the power of relations to affect entities’ behaviors. Indeed, constraints are still perfectly admissible in a pure atomistic worldview where extrinsic relations do not change the immutable intrinsic (monadic) properties of all entities, but can cause quantitative changes in their defining properties—mass, pressure, or velocity, for example. Constraining relations are perfect examples of atomistic kind of relations. Even in the mechanistic scenario

of a set of billiard balls colliding to each other, their motions affect each other and the pool table itself constraints the local movement of those balls.

On the contrary, in genuine ontological emergent systems the causal intra-level relations change their relata *qualitatively*, and that is why their intra-level processes of causation perform the so-called DC that many philosophers attribute to wholes as reified entities.

To be sure, I fully acknowledge the existence of bottom-up constraints, as well constraints imposed by systems' states on their constituents' degree of freedom (Kistler 2009: 603).⁶ I just think that there are *also* causal relations able to qualitatively change the relata's behaviors, and that is what the notion of DC ultimately intends to capture.

On the other hand, instead of adopting a hybrid account of inter-level relations as a combination of an inter-level causal relation with a non-causal relation—either Craver and Bechtel's relation of constitution, or Kistler's relation of constraint—I argue for a straightforward relational-horizontal perspective, conceiving both UC and DC as cases of *intra-level* causal relations seen from the point of view of a joint operation between the individual behaviors of the relata of a given relational system (UC) and the transformative and coercive network relations horizontally distributed across the relational system (DC).

Instead of saying that an entity is a constituent of a whole we should say that the individual is causally related to, being co-dependent of, others entities as *relata* of the same relational domain (call it a system, a whole, or a relational totality). To say that we “ascend” in the vertical compositional hierarchy of wholes (wholes of wholes, etc.) is just a metaphorical way of saying that the *boundaries* of the causal relational environment of a given entity or set of entities are successively being extended. So, instead of adding more floors to our hierarchical-vertical view, we are *horizontally extending* the different types of relational domains. Atoms interacting in specific types of relations with other atoms form a relational system called molecule, molecules interacting in specific types of relations with each other and with other types of relata form a relational system called cell; trees causally interacting in certain ways with each other and with other types of relata form a relational system called forest. There is no need to suppose the existence of wholes as entities existing on upper levels.

In sum, to see causal relations among different elements that share the same relational system into the light of a vertical model of representation of part and wholes is, I argue, a mistake, and it helps to generate the famous *vexata quaestio* of DC and UC. We must distinguish between two different types of relations and their correspondent different models of representation:

- the relation of mereological parthood of ‘*being part of*’ of a whole is compatible with a vertical and hierarchical model of representation;

⁶ As an example of a bottom-up constraint, Kistler mentions the case, shown by Shapiro (2005), that “the structure of the brain, shaped by phylogenetic and ontogenetic evolution, constrains cognitive functions, and thus sets limits on the multirealizability of those functions” (Kistler 2009: 608, n. 15). On the notion of DC as a power of constraining, see Deacon 2003; Thompson 2007: 424–428; Bishop 2008; Hooker 2013.

- the relation of interconnectedness of ‘*being a relatum of*’ another relatum or other relata is instead better represented in a horizontal model of representation.

When we want to consider the causal relations between parts of wholes or between parts and wholes we must thus move from a vertical-mereological perspective to a *horizontal-interconnectedness* viewpoint.

4 The Overcoming of Kim’s Atomistic Metaphysical Principles

In light of what I have been arguing, since there are not two distinct causal processes—one going upwards and the other going downwards—it is clear that we can refuse Kim’s allegation of an overdetermination in cases of DC. There is only *one* complex relational-systemic causal process that can be analyzed through the conceptual distinction between two complementary types of changes, UC referring to the set of the individual causal contributions of each relatum to the overall behavior of their relational system (whole), and DC referring to the way that each relatum is qualitatively affected by all other relata’s actions and interactions.

Kim’s causal closure principle of the physical or of any low-level phenomena is also immediately deniable, as long as we refuse the atomistic metaphysics that Kim adopted—according to which any entity is endowed with a set of absolutely intrinsic and qualitatively immutable defining properties, being able to vary only quantitatively.

According to the atomistic model of part-whole microphysicalism, all systemic properties and behaviors must be explained in terms of the micro-level laws of the system’s constituents. In this sense, a systemic property is non-emergent if it can be fully explainable in terms of (1) an exhaustive knowledge of the intrinsic properties of the lower-level components, in addition to (2) the lower-level laws that obtain over those intrinsic properties (laws concerning the behavior the component parts considered by themselves, and thus considered in a ideal isolation); and to (3) some laws of composition and quantitative interactions—that is, laws that have as their relata the intrinsic properties of the component parts, accounting for their quantitative co-variation, and expressing how the behaviors of the separate parts are compounded when they act together in some proportion and arrangement (Hüttemann 2004: 35; Garson 2006: 230).

On the contrary, from a relational ontology’s viewpoint entities that pertain to low-levels of composition and organization can be qualitatively influenced by entities pertaining to upper-levels of composition and organization, because their *causal relation* is not only qualitatively transformative but it is also an intra-level and horizontal kind of relation. We can say that a molecule is causally influenced and changed by a cell, in the sense that a molecule (as we saw in our graph representation) is integrated in a web of co-dependent transformative and coercive relations with other molecules, ribosomes, and proteins. It is through the expansion of the relational context’s boundaries of a given entity that new levels of organization can be generated. So, while we can talk in terms of lower and upper

levels when we consider the degrees of composition of the different entities, we should talk in terms of relational domains and their horizontal boundaries when we consider the causal relations among those entities.

If entities as relata of different relational domains can influence and qualitatively change each other, one will get a completely different picture from the one given by Atomistic metaphysics. In fact, if entities can qualitatively change or influence each other through their own relations, they will exhibit new relational properties and consequently new patterns of causal behaviors able to be described by new laws or law-like generalizations. Therefore, the causal closure principle can be abandoned as long as there can be different laws that relate different properties exhibited at different domains of relations that correspond to different levels of composition and organization.⁷ No relational domain or level of organization must be conceived as absolutely closed and self-determinable by its own laws, if we admit the possibility that its entities can manifest different properties when they are integrated in different types of relational contexts and levels of organization.

In this sense, there is not only compatibility, but also complementarity between the different nomological levels of organization of reality if the entities that belong to each level are not regarded as self-sufficient essences or as qualitatively immutable atoms whose identities are previous to and independent of their various relational contexts. As Robert Cummings has noted, laws of the so-called special sciences (such as Psychology or Geology) are laws *in situ*, that is, “laws that hold of a special kind of system because of its peculiar constitution and organization. The special sciences do not yield general laws of nature, but rather laws governing the special sorts of systems that are their proper objects of study. Laws *in situ* specify effects—regular behavioral patterns characteristic of a specific kind of mechanism” (Cummings 2000: 121).⁸

If we re-equate DC in the way that we proposed, that is, in a pure relational and horizontal way, and necessarily involving the qualitative influences and changes of

⁷ As Max Kistler observes, “[t]he success of ethology and psychology in explaining numerous animal and human behaviors shows that animals and humans obey “system laws” constraining their evolution at the level of systemic properties, such as cognitive laws determining actions on the basis of reasoning and decision-making. The fact that an organism obeys such laws means that its evolution obeys constraints at a psychological level. The constraints exercised on the organism by laws at different levels, at the level of the organism as a whole, and at various lower levels corresponding to its parts, create no conflict. If the determination of a molecular event is incomplete at its own level, it may nevertheless be completely determined jointly by laws at molecular and system levels. A given molecular event happening in an organism may be partly determined by constraints at the molecular level and partly by downward constraints from the psychological level, insofar as the organism obeys psychological laws” (Kistler 2009: 606).

⁸ According to Cummings, “what grounds the idea that psychology and geology are not fundamental sciences is the thought that psychological and geological systems are special. The principles of psychology and geology and the other so-called special sciences do not govern nature generally, but only special sorts of systems. (...) At the level of fundamental physics, laws are what you get because, at a fundamental level, all you can do is say how things are. We don’t think of the fundamental laws of motion as effects, because we don’t think of them as specifying the behavior of some specialized sort of system that behaves as it does because of its constitution and organization. The things that obey the fundamental laws of motion (everything) do not have some special constitution or organization that accounts for the fact that they obey those laws. The laws of motion just say what motion is in this possible world” (Cummings 2000: 121–122).

the relata through their own intra-level relations, we can realize that the so-called DC is nothing more than the key-mechanism that generates the emergent systemic phenomena—that is, the qualitative changes that entities undergo through their interactions within some relational domains (see Santos 2014). Therefore, there are no two types of processes, one generating emergent phenomena, and the other, downwardly causing or affecting the emergence base. We have one same causal process seen from two points of view: the qualitatively transformative interactions among the individual relata that generate emergent phenomena at the level of the system taken as a whole are the very same interactions that change, constraint and influence each relatum.

5 Conclusion

There is a general consensus that parts cause their wholes upwardly, the only debatable issue being whether wholes can also send causal messages *downwardly* to their parts. According to a widely received view, the emergence base is constituted by entities conceived as endowed with a set of absolutely intrinsic and qualitatively immutable defining properties, their relations being thus conceivable as mere arrangements or combinations, quantitatively affecting only those intrinsic properties. In the light of this atomistic view, we have no option but to conceive wholes as reified entities existing literally above the set of their components and relations, changing them downwardly.

As I tried to show, although this vertical and hierarchical picture can be seen as an adequate model of representation of different levels of composition and organization, it should be discarded as a complete misleading representation of what is really going on with *causal relations* between entities as relata or interactants within their relational contexts.

Assuming a relational-horizontal view of exclusively intra-level causal processes, I argued that we can avoid the idea of cross-level causes, and thus overcome all the problems commonly associated with DC and all the unnecessary confusions and pseudo-metaphysical paradoxes that such type of causal relation raises.

A whole is nothing but an horizontally organized network of agents interacting with each other, as any graph representation can easily show. A whole is just a word that names the overall relational domain that relata and their local relations reciprocally construct, not a new entity existing in a higher level interacting with its parts that lie in lower levels. The only causal agents are the interactants themselves and their interactions. From a relational viewpoint, either you consider the relational system taken as a whole when you *zoom out*, or you consider just one of its relata or some of its sub-relational domains when you *zoom in*.

Therefore, if entities' behaviors are changed and influenced, this happens thanks to *their own intra-level* interactions. As long as those entities are causally connected, their qualitative changes and influences will be propagated and distributed across their entire relational domain. UC and DC are just two complementary aspects, separable only conceptually, of one single ongoing intra-

level complex causal processes among different interactants within a common relational system.

In sum, we can stop invoking wholes as reified and powerful entities existing by themselves in high levels, as we can stop assuming the existence of causes crossing vertical levels, “going up” and “going down”, without, at the same time, losing the deepest ontological significance of the main idea of DC, if we assume a genuine relational ontological perspective as an alternative view to the widespread (although not always conscious) atomistic picture of reality.

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